

IET, DAVV, Indore
B.E. II (Civil)
Class Test II (October-2023)
3AVRC1: Applied Mathematics-III

Time: 70 min

Max Marks: 20

Note: Attempt any FOUR questions. All questions carry equal marks.

Q.1. Solve by Gauss-Seidel iteration Method: (05)

$$27x + 6y - z = 85$$

$$6x + 15y + 2z = 72$$

$$x + y + 54z = 110$$

Q.2. Solve $\frac{dy}{dx} = \frac{1}{x+y}$ for $x = 0.5$ by using Runge- Kutta Method with $x_0 = 0, y_0 = 1$ (Take $h = 0.5$). (05)

Q.3. Use Picards Method to approximate the value of y when $x = 0.1$, given that $y = 1$ when $x = 0$ & $\frac{dy}{dx} = 3x + y^2$. (05)

Q.4. The Area 'A' of a circle of diameter 'd' is given for the following values (05)

d	80	85	90	95	100
A	5026	5674	6362	7088	7854

Calculate the Area of circle of diameter 105.

Q.5. A curve passes through the point (0, 18), (1, 10), (3, - 18) & (6, 90). Find the slope of the curve at $x = 2$. (05)
